

# Design FMEA (qualitative, simulation-signal based) — t4\_r3

Case t4\_r3 | 2026-08-26 | generated by bda (headless) | PDF release

## DFMEA — t4\_r3

| Failure Mode      | Failure Effect                        | Failure Cause (Failure Mechanism)                                                                              | Severity S | Occurrence O  | Detection                                                     |
|-------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------|------------|---------------|---------------------------------------------------------------|
| Failure Mode (F1) | 4C Failure Mode<br>Failure Effect 0 V | anode_potential_v Failure<br>+0.0155 V (< 0 Failure); Failure<br>~15 mV, h=40<br>Failure (R5 V13:<br>-0.417 V) | High       | Medium        | Failure h=50 Failure;<br>Failure/Failure/<br>Failure; Failure |
| Failure Mode      | Failure Mode<br>Failure               | T_max 330.45 K<br>vs 333.15 K Failure,<br>Failure +2.7 K                                                       | High       | Low           | Failure (h<br>Failure ~50<br>Failure);<br>Failure             |
| Failure Mode      | 4.2 V Failure                         | Failure<br>Failure; Failure DFT<br>HOMO/IE-EA<br>Failure (real_co<br>mpute=false)                              | Medium     | Low           | Failure 4.2 V; Failure<br>Failure                             |
| Failure Mode      | Failure Mode<br>Failure               | 1C 5.0383 Ah vs<br>Failure 5.0 Ah                                                                              | Medium     | Low           | Failure/Failure<br>(Failure)                                  |
| Failure Mode      | -20C<br>Failure/Failure               | lowT Failure<br>97.97% (Failure<br>+2.97 pts);<br>Failure<br>(Failure)<br>Failure                              | Medium     | Medium        | Failure;<br>Failure<br>(Failure)                              |
| Failure Mode      | -20C Failure                          | Failure                                                                                                        | High       | Not simulated | Failure<br>-20C Failure;<br>Failure                           |